KALPANA MAMELLA

LinkedIn • GitHub • kalpana150906@gmail.com • 6300969489

Education

Madanapalle Institute of Technology & Science (MITS) | Andhra Pradesh, India
B.Tech, Computer Science & Engineering | 2024–2028 • CGPA: 9.69/10
Sri Chaitanya Junior College | Anantapur, Andhra Pradesh
Intermediate (BIEAP) | 2022–2024 • 97.8%
Jawahar Bharathi English Medium High School | Anantapur, Andhra Pradesh
SSC | 2021–2022 • 91.17%

Technical Skills

- **Languages**: Python, C, Core Java, JavaScript
- **AI / ML**: Machine-learning fundamentals, NLP/semantic similarity fundamentals, computer vision, OpenCV
- **Web Development**: HTML, CSS, JavaScript, React fundamentals
- **Backend & Databases**: FastAPI, SQL, SQLite, MySQL, Firebase
- **Computer Science**: Data Structures, OOP, DBMS, Operating Systems, Computer Networks
- **Tools & Networking**: Git, GitHub, Linux, VLAN concepts, DSCP/DiffServ QoS, VS CODE, CLAUDE CODE, CURSOR, REPLIT

Experience

CODTECH IT Solutions | Web Application Development Intern
- Collaborated on WayFarer Planner, a travel-management web application using HTML, CSS, JavaScript, and Firebase.
- Contributed to responsive user interfaces and interactive front-end components for traveller workflows.

Microsoft Elevate | Project Intern
- Worked on a Driver Drowsiness Detection Alarm System using Python, OpenCV, and Haar cascade classifiers.
- Used facial and eye cues to identify drowsiness indicators and trigger an alert in the prototype.

Prodigy Infotech | Web Development Intern
- Completed front-end development tasks including a responsive landing page, stopwatch, Tic-Tac-Toe game, personal portfolio, and weather application.

Selected Projects

OncoReady AI — AI-Assisted Cancer Care Readiness | Healthcare AI
- Conceptualized a healthcare workflow support system to surface potential care gaps, missing or conflicting clinical-record information, and delays in a patient journey.
- Designed role-specific workflow views for healthcare staff and a simplified patient/caregiver experience; positioned the system as care-coordination support, not a replacement for clinical judgment.

Smart Hospital QoS — Emergency MRI Prioritization System | Computer Networks
- Designed a network QoS concept to prioritize emergency, urgent, and routine MRI-related traffic using DSCP/DiffServ, VLAN segmentation, and priority-queuing concepts.
- Project scope includes network-topology visualization, packet inspection, traffic benchmarking, and QoS monitoring.

AI Recruiter Pro — Candidate Ranking & Matching System | AI / Full Stack
- Developing a resume-to-job-description matching workflow using Python, FastAPI, semantic similarity, and SQLite.
- Planned capabilities include resume parsing, candidate ranking, skill-gap insights, and a recruiter dashboard.

Driver Drowsiness Detection Alarm System | Computer Vision
- Built a prototype using Python and OpenCV to monitor facial/eye cues and trigger an alert when drowsiness indicators are detected.

WayFarer Planner — Travel Management Web Application | Web Development
- Collaborated on a travel-management web application using HTML, CSS, JavaScript, and Firebase; contributed to responsive interfaces and interactive components.

Achievements

- Rashtriya Innovation Challenge 2026 — RITH Tirupati: team selected at institute level.
- Selected at institute level for Smart India Hackathon; participated in SIH 2025 and 2026.
- Secured 1st position in the college Web Development Workshop, GMOCS 2025.
- Participated in ImpactX Hackathon 2025.
- Academic Excellence Award; Pragati Merit Scholarship recipient.

Certifications

- Fundamentals of AI — IBM
- Oracle Certified Foundations Associate
- The Joy of Computing using Python — NPTEL: 84%, Silver badge

Additional

Languages: English, Telugu, Hindi
Interests: Applied AI, healthcare technology, software engineering, problem solving